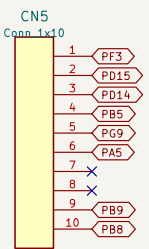
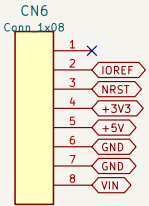


NUCLEO-H563ZI (base board)

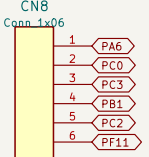
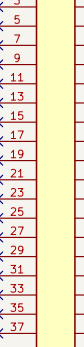
STM32H563ZI - ST Zio / Arduino Uno V3 headers (stack bus source)



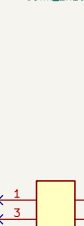
CN7 Conn_2x19



CN10 Conn_2x19



CN10 Conn_2x19



CN10 Conn_2x19



CN7 / CN10 -> H5632 morpho (2x19) pass-through to IKS41.
GPIO/Analog carrier 511 up the stack (DMS115): not used by the ToF/IMU sensor stack -> no-connect here.

Key STM32 functions on the Arduino headers:

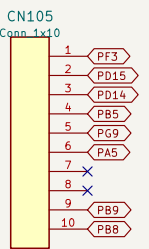
PBB=STCL_SCL, PB9=ICL1_SDA, PBS=TIM3_CH2(CLK_IN),
PBL=SYNCL_M, PBA=XSCHUT, PB7=INTRE(XIT7)

Default jumpers (board config, not on Zio pins):

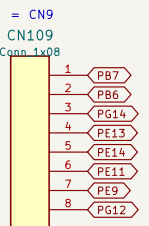
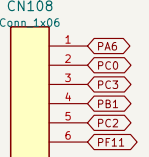
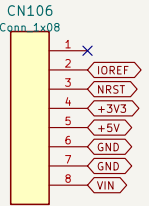
JP2=1-2 (USB/UCPD power) JPA=1-2 (VDD_MCU=3V3)
JP5=closed (I2C measure) JP6=closed (ST-LINK VCP UART)

X-NUCLEO-IKS4A1 (middle board)

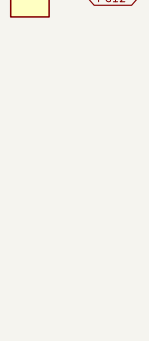
MEMS + environmental. Taps I2C on PB8/PB9 as legacy-I2C targets.



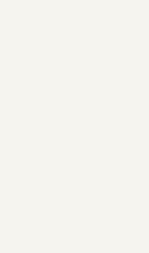
CN107 Conn_2x19



CN109 Conn_1x9



CN109 Conn_1x9



CN109 Conn_1x9

CN109 Conn_1x9

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U1 LDK130: +3V3 -> +1V8 (sensor Vio; JP-selectable to 3V3)

5Vx secondary sensors above share the Internal 1V8 I2C bus (HS_SCL/HS_SDA).
Real board splits it into STM_I2C / SINK_I2C / HUB2_I2C sub-buses via solder bridges, and HUB1/HUB2 can master sensors over their own I2C. INT/DRDY lines are SB-rotatable to Arduino pins (not used as EXTI by firmware) -> NC here.

Default jumpers:

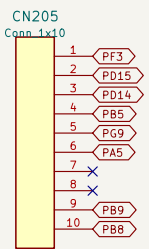
JA/JB = 5-6 (I2C bus routing: STM I2C -> sensors)

J2 = open (USER_INT selector off -> sensor INTs unmasked -> NC)

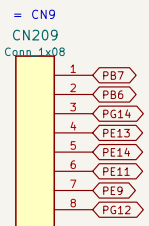
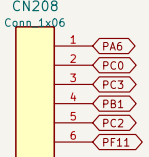
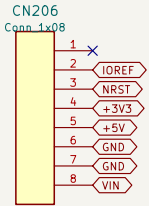
JP1/JP2 = open (Vio / BTLrq selectors) JP5 = 1-2 (DIL24 = 3V3,I0)

X-NUCLEO-53L9A1 (top board)

VL53L9CX ToF 3D LIDAR. I3C on PB8/PB9, level-shifted to sensor IOVDD.



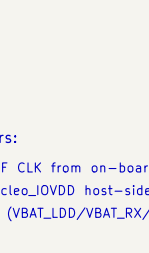
CN206 Conn_2x19



CN209 Conn_1x9



CN209 Conn_1x9



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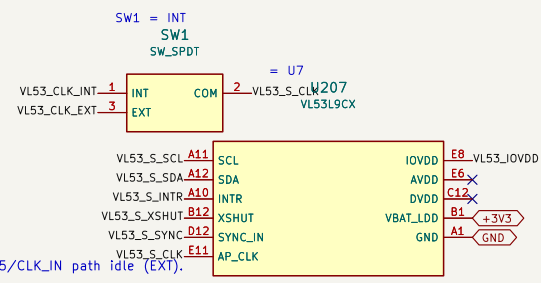
CN209 Conn_1x9

Default jumpers:

SW1 = INT (ToF CLK from on-board Y1 12MHz; see switch)

J1 = 3V3 (Nucleo_IOVDD host-side shifter reference = 3.3V)

J2,J5 = linked (VBAT_I00/VBAT_RX/AVDD/DVDD ratio powered)



U7 VL53L9CX ToF (5x52). On-board LDOs (from CM6 +5V) derive IOVDD/AVDD(QVDD)/DVDD(I1V2) - AVDD/DVDD abstracted (NC).